Communications Scope of Works

Skygate Play Lot SKG26 – B345 & 346

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Brisbane Airport

Document Control

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Contents

1. Introduction

The following BAC Communications specifications are to be used for Lot SKG26 Skygate Play project only

BAC Communications services include network connections for monitoring of the BMS and metering. Provision is also made for carrier cable access to the office area.

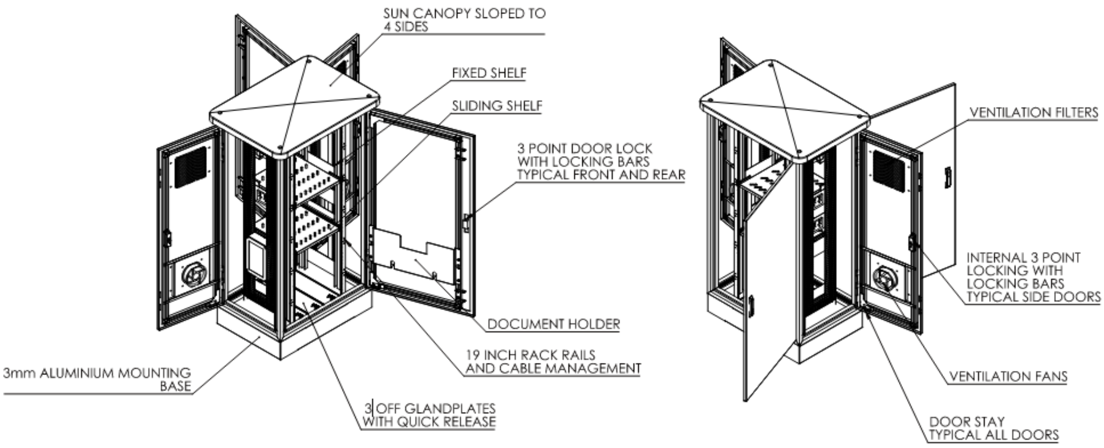
The fibre termination and network equipment will be installed within a separate lockable BAC ICT enclosure within the external Mechanical Plant area and equipment shall be ‘hardened’ to withstand higher temperatures.

Any variations to the requirements within this document require approval from the Airport Communication Systems Manager before installation.

Contact Shaun Strupis (3068 6679) for any questions regarding this document.

2. Communications Distributors

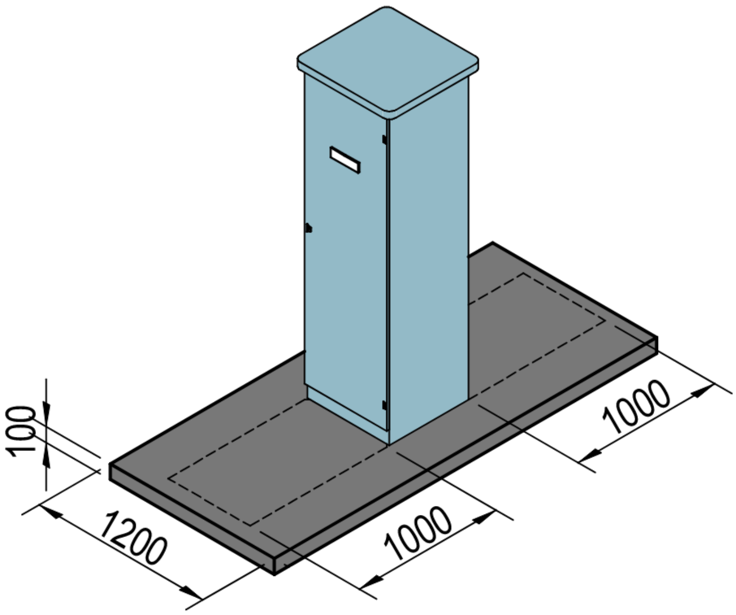
2.1 ICT Distributor Specifications – Field Cabinet



<table>
<tr><th>Supporting Services</th><th>Requirements</th></tr>
<tr><td>Size and Construction</td><td>

To match the requirements of Logix ITCS-BAC-1-45-01 (including provision of internal cable reticulation and shelving)
AS2700:2011 T33 “Smoke Blue” colour
Mounting pad and clearance requirements:

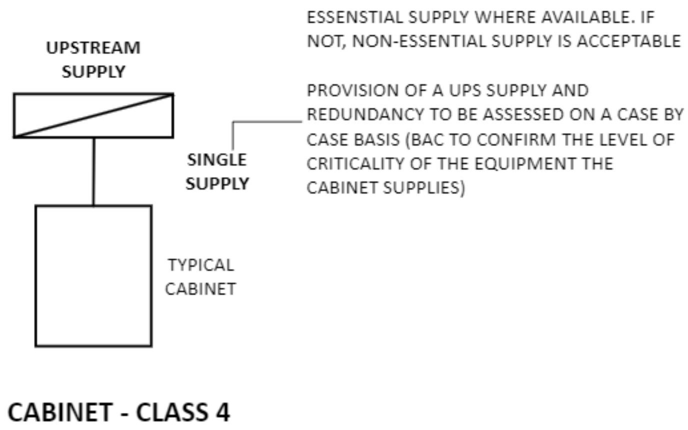
The concrete pad is to sit slightly higher than the finished ground level to prevent water pooling on the slab. The ground surface is to be finished to ensure no trip hazard is created
See Figure 13 for clearance requirements:

Figure 1 Field Cabinet Clearance Requirements


</td></tr>
<tr><td>Power</td><td>

Supply type and level of redundancy to assessed on a case-by-case basis.

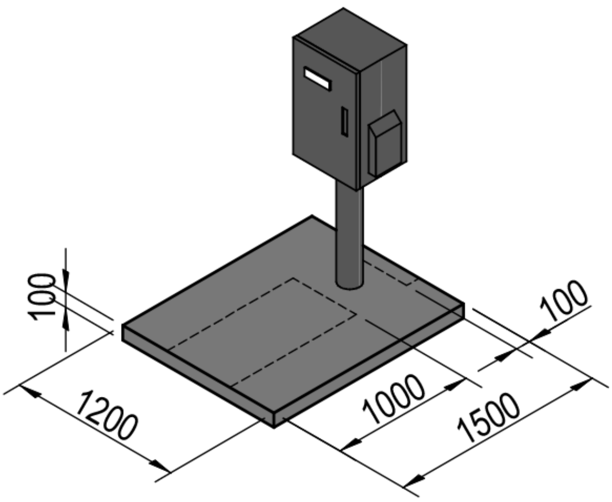
</td></tr>
</table>

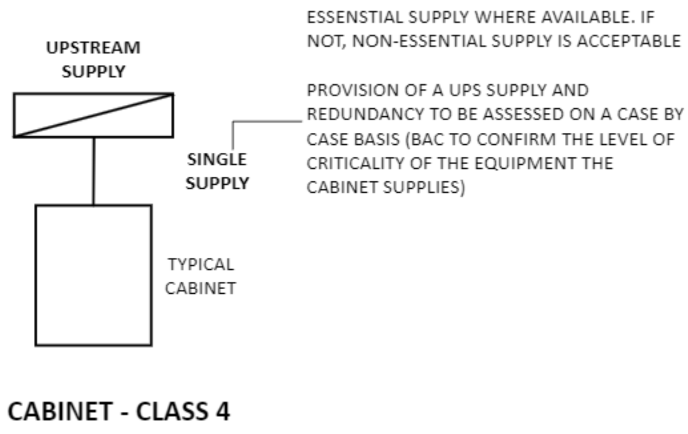
Supporting Services	Requirements
	• See Figure 14 for details: **Figure 2 Field cabinet power supply arrangement** 
Vertical Cable/Path Lead Management	• Vertical patch lead/cable manager shall be provided with front cover • Vertical cable minders shall be installed to side of each rail • Vertical manger shall be positioned so that the front face aligns with horizontal management front face • Model: Rack Technologies IQ series (detailed below) or equivalent approved by BAC's comms infrastructure manager • 45RU part no. IQCM4560
Horizontal Cable/Patch Lead Management	• Horizontal path lead/cable manager shall be provided with front cover • A minimum of 2 x 93mm deep Horizontal patch lead minders shall be installed. Note: More may be needed depending on the quantity of equipment installed • Horizontal cable minders to be 1RU. • Model: Rack Technologies IQ series (detailed below) or equivalent approved by BAC's comms infrastructure manager – 1RU part no. IQOF9509 - 4 off – 2RU part no. IQOF9502 - 1 off
Power Supply and Power Distribution Unit (PDU)	• The cabinet shall be supplied with an integral switchboard • The switchboard shall be labelled with Switchboard and circuit breaker that the cabinet is fed from • The load centre shall be single phase and have 12 circuit breaker positions and have an integrated surge suppression device • One circuit off the load centre to supply a Clipsal switched captive outlet (Clipsal part no. 563C315D) installed m 40mm above the base of the cabinet to provide power for the cabinet • The cabinet shall be serviced by a single 6 or 10-way vertical PDU fed from a captive power outlet
CES	• Earth-Neutral bar in accordance with AS/CA S009 and AS/NZS 3000, with sufficient capacity for direct connection to all applicable elements of the cabinet

Supporting Services	Requirements
Fibre Connection	• Level of redundancy to be assessed on a case-by-case basis, subject to services connected to the cabinet • Where a direct cable route between distributor and end use is not available, patching of backbone connections through distributors shall be used • Note: field cabinets are typically connected through a network ring
Ventilation	• Filtered ventilation using air drawn from outside ambient air • Dual fans, thermostat controlled, pressurising filtered ventilation – Note: Two vent fans with individual thermostat controls to be mounted on the right-hand side pods pressurising air in the cabinet and expelling it through the left-hand side
Fire Services	N/A
Security	• EKA Lock for access control and ability to audit trail • Tri locking Stainless steel lockable handle. Lock barrel to suit BAC's EKA lock system (For lock barrel details – contact the Project Manager) • Cabinet door open monitoring and alarm, with alarm to central system using limit switches for 'door open' alarm • Vehicle barrier – subject to location
Remote Monitoring and Control	• Temperature monitoring • Where UPS supply is required, UPS monitoring
Cable Pathways	• As detailed in design documents
Lighting	• High efficiency LED lights installed at front and rear • Manual lighting control function - Operated by door switch

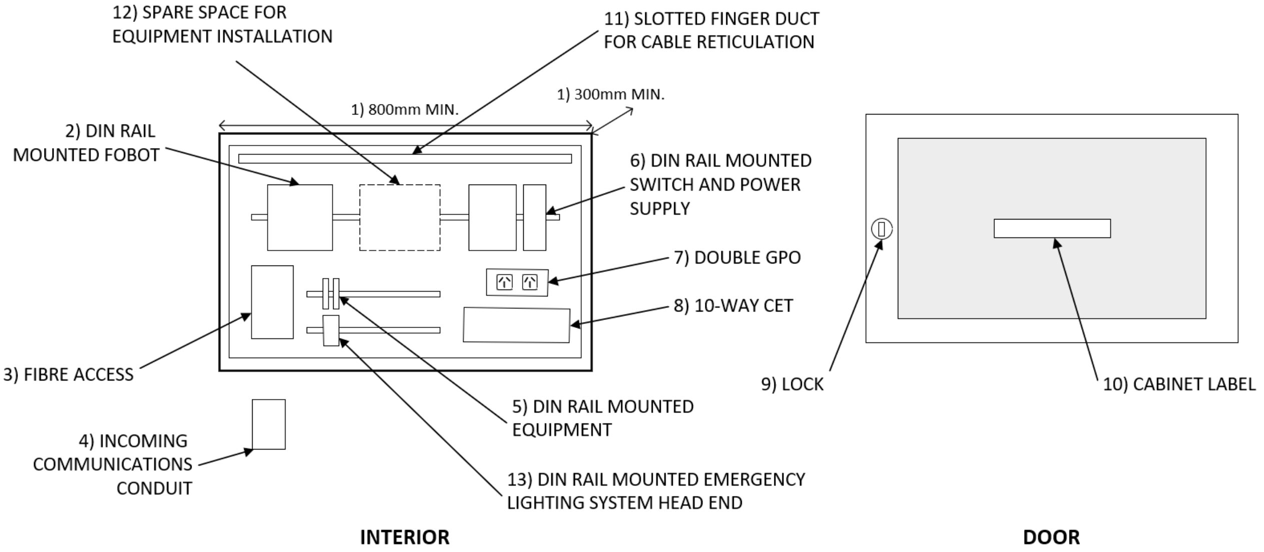
2.2 ICT Distributor Specifications – Pedestal Cabinet

Images for Outdoor cabinet installation

Supporting Services	Requirements
Size and Construction	• To match the requirements of Logix PME Enclosure (PME100603_S-01-BAC) • Marine grade stainless steel colour • Mounting pad and clearance requirements: – The concrete pad is to sit slightly higher than the finished ground level to prevent water pooling on the slab. The ground surface is to be finished to ensure no trip hazard is created – See Figure 13 for details: **Figure 3 Field Pedestal Clearance Requirements** 
Power	• Supply type and level of redundancy to assessed on a case-by-case basis

Supporting Services	Requirements
	• See Figure 17 for details: **Figure 4 Field Pedestal power supply arrangement** 
CES	• Earth-Neutral bar in accordance with AS/CA S009 and AS/NZS 3000, with sufficient capacity for direct connection to all applicable elements of the cabinet
Fibre Connection	• Level of redundancy to be assessed on a case-by-case basis, subject to services connected to the cabinet • Where a direct cable route between distributor and end use is not available, patching of backbone connections through distributors shall be used • Note: field cabinets are typically connected through a network ring
Ventilation	• Filtered ventilation using air drawn from outside ambient air • Dual fans, thermostat controlled, pressurising filtered ventilation – Note: Two vent fans with individual thermostat controls to be mounted on the right-hand side pods pressurising air in the cabinet and expelling it through the left-hand side
Fire Services	N/A
Security	• EKA Lock for access control and ability to audit trail • Tri locking Stainless steel lockable handle. Lock barrel to suit BAC’s EKA lock system (For lock barrel details – contact the Project Manager) • Cabinet door open monitoring and alarm, with alarm to central system using limit switches for ‘door open’ alarm • Vehicle barrier – subject to location
Remote Monitoring and Control	• Temperature monitoring • Where UPS supply is required, UPS monitoring
Cable Pathways	• As detailed in design documents
Lighting	N/A

2.3 ICT Distributor Specifications – Warehouse Cabinet



Item	Requirements
Warehouse Communications Cabinet	• Minimum cabinet dimensions: 800x500x300 (w)x(h)x(d) • Din rail mounted FOBOT • Fibre cable access from pit to FOBOT • Incoming communications conduit shall align with the left-hand side of the compartment to allow the fibre to feed directly into the FOBOT • Din rail mounted data cable terminations • Din rail mounted switch and power supply • One double GPO (Outlet to be labelled with DB and C/B) • 10-way CET within the comms compartment • The comms compartment door shall have an EKA lock (enrolled on the BAC EKA system) • A stainless-steel label shall be engraved with the cabinet ID and Name (as supplied by BAC IT) secured to the comms compartment door • Slotted finger conduit for cable reticulation • Spare space on din rail for future installation • Din rail allowance for Emergency lighting system head end to be mounted on. • Power supply, including UPS provision to be assessed on a case-by-case basis • Refer to Figure 18 for cabinet set out
External Cable Pathways and Pits	• Pit and conduits shall be installed per the requirements set in section 0of this design guideline • Cable pathways from the comms panel shall be installed to facilitate a connection to the electrical metering panel / water meter
Fibre installation	• Minimum 12 core OS2 SM fibre

Item	Requirements
Data and Cable Installation	• 1 x single data to metering panel for Moxa serial/ethernet converter • Patch leads can be run to adjacent compartments • Where the cable run is via underground conduit, cables must be underground rated
UTP Patch Leads	• Provide UTP patch leads with the following features and quantities: – Patch leads shall be Panduit TX6A™ Category 6A Performance 28 AWG Shielded Patch Cords or approved equivalent 4 pair stranded conductors. – Factory terminated RJ45 modular plugs – AS/CA S008 compliant – Sufficient Quantity and lengths to connect all BAC devices to the network switch and network devices to field outlets
Labelling	• Labels shall be applied to all cabinets, FOBOTS, patch panels, cables and equipment installed • Labels shall comply with current version of BAC's ICT Cabling Installation Specification • All equipment shall be labelled • Comms Infrastructure label names shall be requested from BAC's comms infrastructure manager
Network Equipment	• Network equipment shall be based on the BAC Supported Network Hardware V2

2.4 ICT Distributor Locations

Locations for Communications Distributors are to be carefully considered as once installed, they are costly and disruptive to relocate. Criteria to be considered include.

Communications Distributor location selection criteria	Supporting information
Security of tenure	Where possible, Communication Distributors need to be located where they are unlikely to be relocated. Relocation of communications Distributors is expensive and disruptive to operations
Landside or airside	• Landside infrastructure to originate from ICT Distributor on landside • Airside infrastructure to originate from ICT Distributor on Airside
Away from public access	To reduce the risk of accidental damage or vandalism
Proximity to field equipment serviced by the ICT Distributor	Maximum 90 metre category 6A permanent link (to cover all vertical and horizontal transitions) between the communication distributor and field equipment outlet. This is to guarantee the data cabling meet the performance standards
To be located away from EMI sources	Such as: • Lifts • High voltage sources • Transformers • Baggage belts or other equipment with motors
To be located away from wet areas	To prevent operational disruptions and physical damage from leaks or condensation, it's recommended to refrain from situating communication rooms beneath moist environments like changing rooms or toilets
To be located away from water ingress	• Water drainage paths must not be channelled above or within room • Not Prone to stormwater or flooding (for outdoor cabinets)
Serving other levels	Distributors shall only service the area on a single floor unless approved by BAC's comms infrastructure manager
Sufficient space leading to the room for equipment relocation	Full height communication racks typically have a height of 2.2m. and are supplied as a single unit. Consideration be given to the space allowed for relating equipment to the communications room
Protection against damage	Distributors shall be properly safeguarded against potential damage that could reasonably occur due to exposure to weather, water, excess moisture, corrosive fumes, dust buildup, steam, oil, high temperatures, or any other conditions they may encounter during normal use. Additionally for external distributors, appropriate physical barriers such as bollards or barriers should be used where necessary to prevent accidental impacts or collisions.

2.5 Site Specific ICT Distributors

2.5.1 Tenancy 1 - Building 346 BAC ICT Cabinet

Distributor Type – Warehouse cabinet

2.5.2 Tenancy 2 - Building 345 BAC ICT Cabinet

Distributor Type – Warehouse cabinet

3. External Cable Pathway and Pits

Cable pathway and pit installation shall comply with ADG-701 & ACG-701

3.1 BAC Lead-in Pit

A new communication services pit shall be installed over the communications duct bank in the services corridor at a suitable location agreed between BAC's comms infrastructure manager and the developer
If a pit already exists outside the site, communications services shall be delivered from that pit.

Where BAC communications infrastructure enables supply to the site from 2 different directions, and, where required by the building owner, a diverse path feed to the building is achieved by establishing a second BAC pit over the BAC conduit bank, a second property pit, lead-in conduit etc. The second BAC pit should be a minimum of 10 metres away from the first

3.2 Property Pit (Building Entry Pit)

The building developer shall supply and install a property pit inside the site boundary. Position to be determined based on-site requirements, catering for connectivity to both BAC communications enclosure and tenant communications room. This pit will be the connection point for underground conduits from either the BAC or Carrier networks servicing the site

It is recommended that the property pit and the building entry pit are lockable to protect the Communications cables from interference

3.3 BAC Lead-in cable pathway

Underground lead-in conduits shall enter the Building Distributor via property pit.
The Property pit shall be positioned no more than 15m from the building entry point.

The building owner/developer shall provide 2 X 100mm communications conduits from the BAC Lead-in pit to the property / building entry pit

A minimum of 1 x 100mm comms conduit shall be installed between the building entry pit and the MDF/BD room and a minimum of 1 x 100mm comms conduit shall be installed between the building entry pit and the BAC communications cabinet

Where building entry conduits appear in the building in a location other than the MDF/BD cable tray shall be installed from the building entry to the location of the BAC infrastructure

3.4 Site Specific Information

There are two existing street pits available to access conduit within this area, they are to be utilised for conduit connection. (The existing pit at the Boulevard end will be installed as part of the substation project)

To meet BAC standards providing site pathway for communications infrastructure this work should be installed as per diagram in Appendix E including: -

- Building 346 Tenancy 1
    - Existing comms street pit FCP BVDxx is to be installed and named as part of the substation project.
    - 1 x Cubis C2 Belltop 8 SL1 Pit for the building entry pit.
    - 2 x 100mm communications conduit to be installed from existing street pit FCP BVDxx to building entry pit.
    - Install 1 x 100mm comms conduit from building entry pit to BAC IT comms distributor.
    - Install 1 x 100mm comms conduit from building entry pit to B346 comms room for NBN / Carrier cable installation.
    - Conduits entering the building to be sealed using the appropriately sized fire bags and fire seal mastic, configured to enable cable access for future use.
    - Conduits to have bell mouths installed into pits sealed against pit wall with mastic or similar – expanding foam not to be used.
    - Other conduits may be required to facilitate the data cabling for the MSB metering, Emergency lighting, future solar and MSSB BMS network connections. There is a 90m distance limitation on data cabling, confirm distance is within spec otherwise additional comms cabinets, fibre, power and active network equipment may be required.
- Building 345 Tenancy 2
    - Existing comms street pit FCP CIR51
    - 1 x Cubis C2 Belltop 8 SL1 Pit for the building entry pit.
    - 2 x 100mm communications conduit to be installed from existing street pit FCP CIR51 to building entry pit.
    - Install 1 x 100mm comms conduit from building entry pit to BAC IT comms distributor.
    - Install 1 x 100mm comms conduit from building entry pit to B345 comms room for NBN / Carrier cable installation.
    - Conduits entering the building to be sealed using the appropriately sized fire bags and fire seal mastic, configured to enable cable access for future use.
    - Conduits to have bell mouths installed into pits sealed against pit wall with mastic or similar – expanding foam not to be used.
    - Other conduits may be required to facilitate the data cabling for the MSB metering, Emergency lighting, future solar and MSSB BMS network connections. There is a 90m distance limitation on data cabling, confirm distance is within spec otherwise additional comms cabinets, fibre, power and active network equipment may be required.

Refer to Annexure 'C' for details of permissible pit types

4. Cable Pathway

Installation shall comply with ADG-701 & ACG-701.

Cable pathways from the comms panel must be installed to facilitate a connection to: -

- The Air conditioning plant (for BMS monitoring)
- The electrical metering panel / water meter

5. Fibre Optic Installation

Must comply with the current version of ADG-701 & ACG-701.

Supply and install: -

5.1 Fibre Optic Lead-in cable for Tenancy 1 – Building 346

- 12 Core OS2 SM Loose tube fibre to be installed from new site substation to B346 BAC ICT Comms distributor
- Install an AFL Mini DIN rail mounted FOBOT fibre enclosure in new substation.
- Install an AFL Mini DIN rail mounted FOBOT fibre enclosure in B346 BAC ICT Comms distributor

5.2 Fibre Optic Lead-in cable for Tenancy 2 – Building 345

- 12 Core OS2 SM Loose tube fibre to be installed from CICCT-CDG-CP07 to B345 BAC ICT Comms distributor
- Install in existing AFL fobot in CICCT-CDG-CP07
- Install an AFL Mini DIN rail mounted FOBOT fibre enclosure in B345 BAC ICT Comms distributor

6. Copper SCS Installation

Must comply with the current version of the ADG-701 & ACG-701.
Install: -

- Cabling manufacturer must be a BAC approved vendor
- All copper SCS cables shall be terminated on din rail mount outlets at both ends of the permanent link.
- 1 x single C6A F/UTP data outlet to the air conditioning control panel for each mechanical services network device (Mech Services Contractor to provide quantities). Please confirm quantities.
- 1 x single C6A F/UTP to emergency lighting controller (If not located in the BAC ICT cabinet)
- 1 x single C6A F/UTP to metering panel for Moxa serial/ethernet converter.

- 1 x single C6A F/UTP to future solar location
- Patch leads can be run to adjacent compartments.
- Where the cable run is via underground conduit, cables must be underground rated.

7. Patch Leads

Must comply with the current version of ADG-701
Provide UTP patch leads with the following features and quantities:

- Patch leads shall be PoE++ compliant.
- Patch leads shall be Panduit TX6A™ Category 6A Performance 28 AWG Shielded Patch Cords or approved equivalent.
- Patch lead lengths shall be selected to ensure that minimum slack needs to be accommodated within cabinet patch lead management systems
- Sufficient Quantity and lengths to connect all BAC devices to the network switch and network devices to field outlets.

8. Patched Fibre Services

All fibre test results are to be approved by BAC before fibre services are patched.

Fibre patching to be performed by BAC nominated contract Programmed Electrical Technologies (refer to TS-4451 - ICT - System Works Specification).

Fibre services connect the switch into REP Ring 117. Refer Appendix D & E.
Scope includes: -

- Install new fibre service CSBNE-FSLB1990 from BACS1-R117-B346-01 to BACS1-R117-B345-01.
- Install new fibre service CSBNE-FSLB1990 from BACS1-R117-CP02-01 to BACS1-R117-B346-01.
- Amend fibre service CSBNE-FSLB646 to connect to BACS1-R117-B346-01.
- Supply all single mode patch leads.
- All connectors to be cleaned and scoped to confirm interfaces are clean.
- Leads to be labelled with relevant service number at both ends.
- Entire service (including patch leads where applicable) to be tested by laser torch.
- Patch sheets to be updated with cable IDs and port numbers (and any other relevant information).
- Patch sheets to be returned to Shaun Strupis in in Word and PDF format (together with LSPM and OTDR results)
- Base patching sheet to be supplied by BAC (Shaun Strupis)
- Example of BAC Fibre patching Schedule

Schedule 3 - BAC Fibre Optic Service Patching Record

Customer:	BAC IT	Date Patched:	Amended 11.07.25	Service Number:	CSBNE-FSLB1234
Service Type:	SM	Number of cores:	2	Service Purpose:	Rep 142 – B302-06 (Aus Post) to B306 (Aramex)
'A' End Location (From)	B302-06				
'B' End Location (To)	B306 (Aramex)		Maximo W/O:		

Notes:

a. Testing to be performed in accordance with BAC Fibre Optic Installation and Testing Requirements document.
b. BAC Budget Loss calculation and Result Summary sheet to be completed.

Laser Only

		Patch From				Patch To			
Item	Location	From Cabinet ID	Cable ID	Port (or Core) Number	Connector type (or splice)	To Cabinet ID	Cable ID	Port (or Core) number	Connector type (or splice)
1	B302-06	A6	'A' End Equipment: (BACS1-R142-B302-06 1/2) 2m SCA-LC Patch Lead (SM)			A6	CIBOR-FS2-B302-02-01 6 core OS2 from B302-02 to CDG-B302-06	03:04	SCA
2	B302-02	A1	CIBOR-FS2-B302-02-01 6 core OS2 from B302-02 to CDG-B302-06	03:04	SCA	A1	CIBOR-FS2-B302-A1-01 6 core OS2 from B302-A1 to B302-A2	03:04	SCA
3	B302-01	A1	CIBOR-FS2-B302-A1-01 6 core OS2 from B302-A1 to B302-A2	03:04	SCA	A1	CIBOR-FS2-S481-A1-01 12 core OS2 from S481 to B302-A1	01:02	SCA
4	S481	A1	CIBOR-FS2-S481-A1-01 12 core OS2 from S481 to B302-A1	01:02	SCA	A1	COBOR-FS2-S479-A1-01 12 core OS2 from S479-S481	05:06	SCA
5	S479	A1	COBOR-FS2-S479-A1-01 12 core OS2 from S479-S481	05:06	SCA	A1	CIBOR-FS2-S479-A1-02 12 Core OS2 from S479 to B306	01:02	SCA
6	B306 (Aramex)	A1	CIBOR-FS2-S479-A1-02 12 Core OS2 from S479 to B306	01:02	SCA	A1	'B' End Equipment: BACS1-R142-B306-01 Gi 1/1 (device name and port) 2m SCA-LC Patch lead (SM)		

Other Details

Length (from LSPM)		DB Loss Core 1 (show both wavelengths)		DB Loss Core 2 (show both wavelengths)	

9. Communications Labelling

Labels shall comply with current version of ACG-701

- Labels shall be applied to all cabinets, FOBOTS, patch panels, cables and equipment installed
- All equipment shall be labelled
- Comms Infrastructure label names shall be requested from BAC's comms infrastructure manager

10. Security

Installation shall comply with ADG-701, ACG-701 and DG-4410

10.1 Site Specific security requirements

There is no BAC CCTV or EACS on site.

11. Network Equipment

Refer to the BAC Network Design Guide spreadsheet and follow the instructions in the "Read Me First Tab".

Ports are to be configured for the following: -

- Metering x 1
- BMS x TBA (one port for each mechanical services controller requiring ethernet*)
- Emergency Lighting x 1
- Future Solar

* Information is needed from the mech services controls designer before finalising the number of ports.

Reference must be made to the TS-4451 BAC ICT Systems Works Guidelines to ensure all aspects of BAC requirements are understood (including updating current state diagrams etc). This document lists (among other things) approved contractors permitted to undertake design on the BAC network.

The Bill of Materials must be chosen from the current BAC Approved Network Hardware List.

Network contractor to provide the completed BOM to the BAC Network Architect for approval.

12. Example Bill of Materials for Typical Switch

Line Number	Item Name	Description	Service Duration	Lead Time	Included Item	Quantity
1.0	IE-2000-8TC-G-B	Catalyst IE2000 Switch 8 port PoE 2 x Dual Uplink IP Base	N/A	14 days	No	1
1.0.1	CON-SNT-IE2000GE	SNTC-8X5XNBD POE on LAN base	12 month(s)	N/A	No	1
1.1	PWR-IE65W-PC-AC	PlugPower CordAustralian10A Right Angle	N/A	14 days	No	1
1.2	GLC-LX-SM-RGD=	Cisco Rugged SFP - SFP (mini-GBIC) transceiver module - GigE	N/A	14 days	No	2

Note : This table is provided for information only. **The (BAC Approved) network design contractor must produce a bill of materials detailing equipment complying with BAC Approved Network Hardware list and based on the number of ports required.**

Note: Switch to be sized to provide 20% spare ports at completion.

The BAC IT Change Management process is to be followed before new IT network equipment is connected into the BAC IT operational network.

Connect the switch to the CET.

13. Appendix A - Applicable BAC Specifications and Guideline Documents

All current BAC Guidelines can be found on BNE Community.

Link to complete BAC Guidelines folder.

https://bnecommunity.com.au/media_center/folder/90aa3d81-5547-49ad-9201-6bd69aef517a

Link to BAC ICT guidelines within this folder.

https://bnecommunity.com.au/media_center/folder/f279c8a7-d73e-4330-9415-fe954e37c6b6

14. Appendix C – Specified Pits

Refer to ACD-701 – 8.2.8 Communications Pits

Building/facility entry pits shall be equivalent to Cubis Custom SCEC 911 SL1 access pits.

Element	Part Number
911 SL1 Pit Body	50201059
911 SL1 Secure Lid	50201203

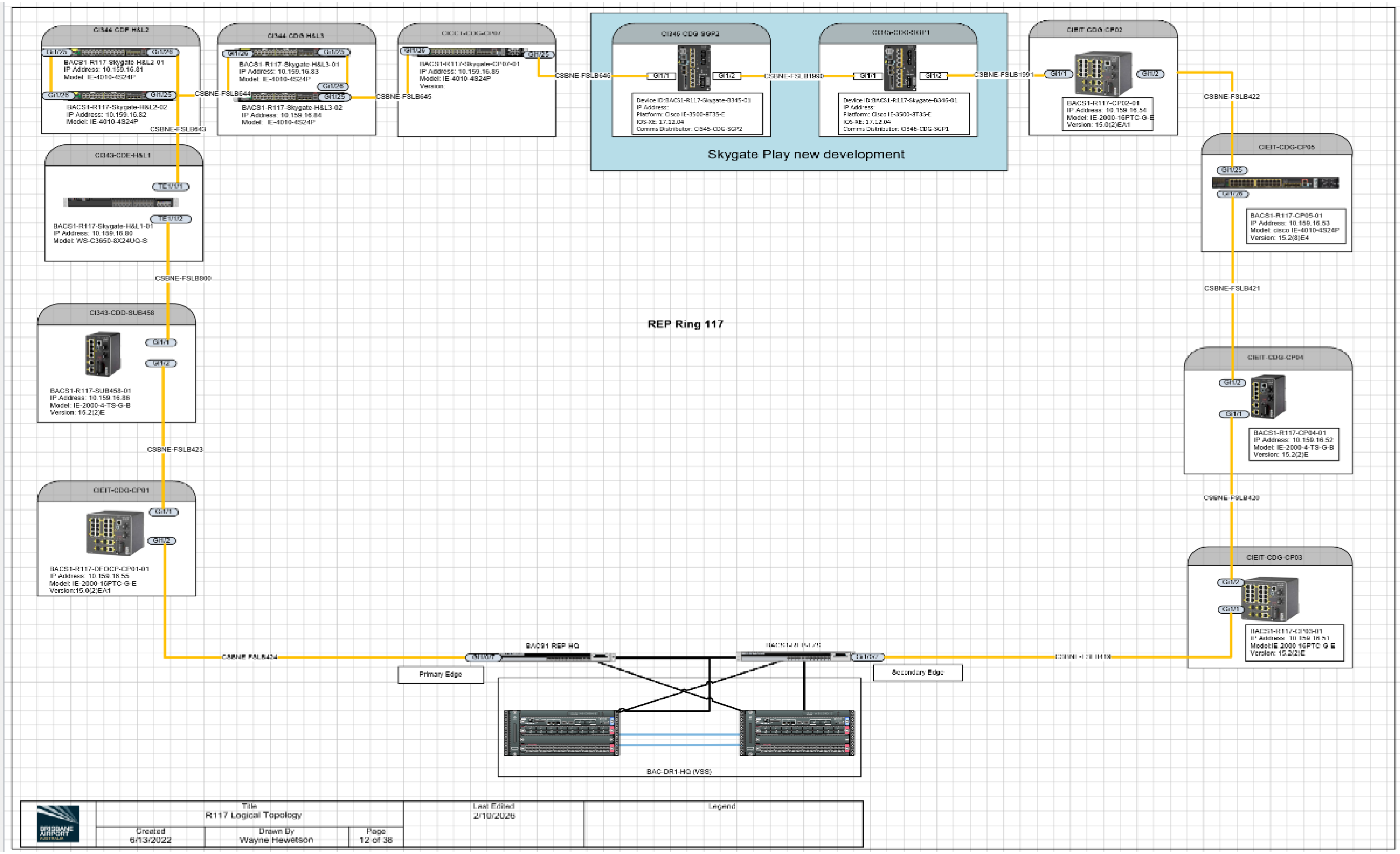
Where a pit is required as a pull point on long conduit runs it shall be equivalent to the Cubis C2 Belltop 8 SL1

Element	Part Number
C2 Belltop 8 SL1 Pit Body	50201060
SL1 Secure Lid	50201144

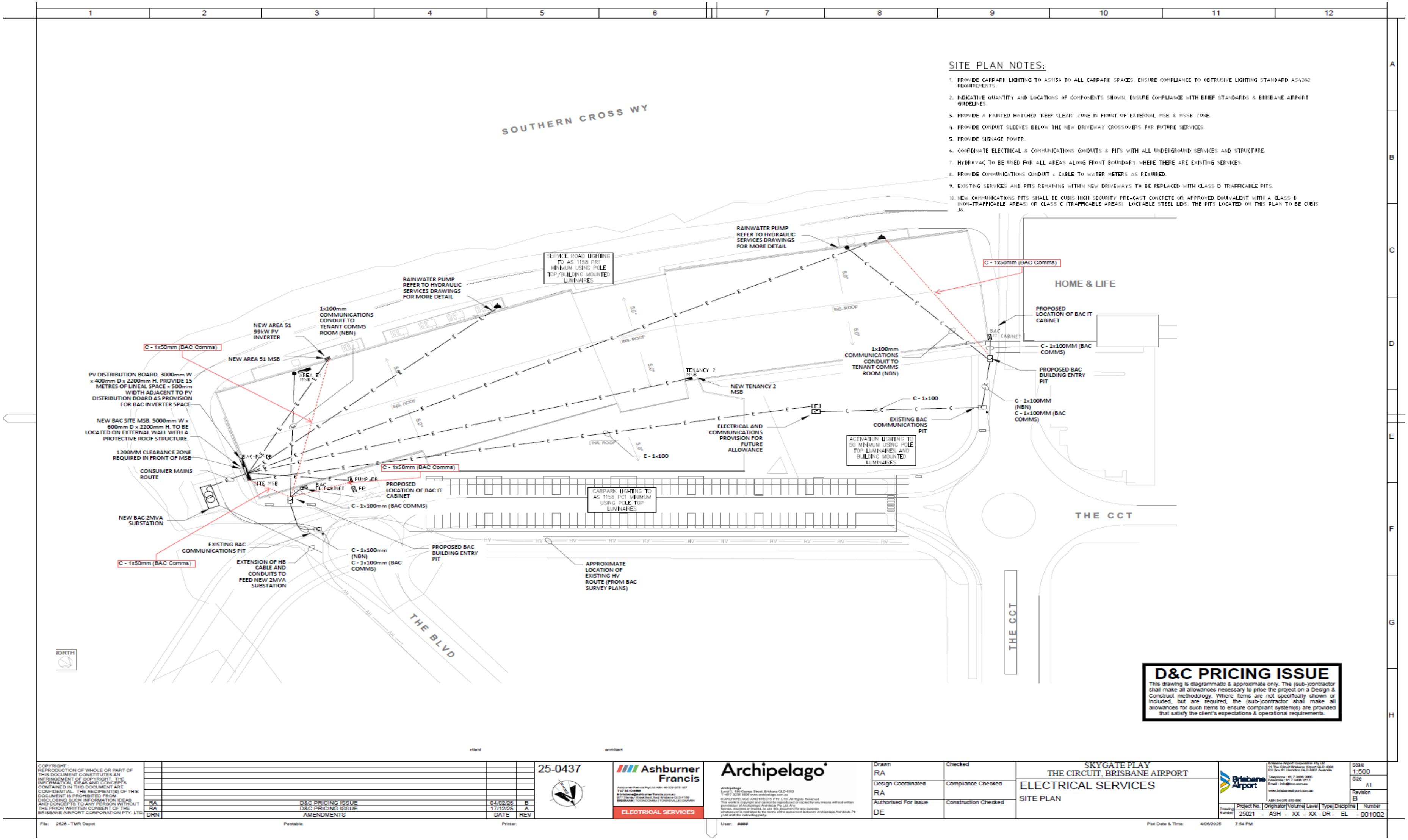
If there are any issues regarding the installation of pits the installation contractor shall advise the BAC's Network and Systems team (or delegate).

Where pits are installed in concreted areas the pit shall be physically tied to the surrounding concreted area to prevent the pit from subsiding below the surrounding area.

15. Appendix D - Network details



16. Appendix E - Site Underground Comms



Preliminary site plan. Not to be used for construction.